IOAA Syllabus

A study edition · MAIS Physics & Astronomy Club

Reformatted from the IOAA 2026 website. Source credits IOAA; updated 17 November 2021. Topic wording is preserved with minor punctuation and typography normalization. Club topic codes and practical-topic labels are organizational aids, not official designations.

Official source: IOAA 2026 syllabus

GENERAL NOTES

1. Extensive contents in basic astronomical concepts are required in theoretical and practical problems.

2. Basic concepts in physics and mathematics at high school level are required in solving the problems. Standard solutions should not involve use of complex numbers and/or solving differential equations.

3. Astronomical software packages may be used in practical and observational problems. The contestants will be informed the list of software packages to be used at least 3 months in advance. The chosen software packages should be preferably freeware or low-cost ones enabling all countries to obtain them easily for practice purpose. The chosen software should preferably be available on multiple OSs (Windows / Unix / GNU-Linux / Mac).

4. Concepts and phenomena not included in the Syllabus may be used in questions but sufficient information must be given in the questions so that contestants without previous knowledge of these topics would not be at a disadvantage.

5. Sophisticated practical equipment likely to be unfamiliar to the candidates should not dominate a problem. If such devices are used in the questions, sufficient information must be provided. In such cases, students should be given opportunity to familiarise themselves with the equipment before the exam.

6. The original text of the problems must use SI units, wherever applicable. Participants will be expected to give appropriate units in their answers and should be familiar with the idea of correct rounding off and expressing the final result(s) and error(s) with the correct number of significant digits.

7. A standard table of constants (supplied as a supplement to this Syllabus) shall be used when setting questions and provided to the students. Additional constants may be added to specific questions as required.

THEORETICAL PART

Symbol (Q) is attached to some topics in the list. It means qualitative understanding only. Quantitative reasoning or proficiency in these topics is not mandatory. The following theoretical contents are proposed for contestants.

BASIC ASTROPHYSICS

01.01 Celestial Mechanics

Newton’s Laws of Gravitation, Kepler’s Laws for circular and non-circular orbits, Roche limit, barycentre, 2-body problem, Lagrange points

01.02 Electromagnetic Theory & Quantum Physics

Electromagnetic spectrum, Radiation Laws, Blackbody radiation

01.03 Thermodynamics

Thermodynamic equilibrium, Ideal gas, Energy transfer

01.04 Spectroscopy and Atomic Physics

Absorption, Emission, Scattering, Spectra of Celestial objects, Doppler effect, Line formation, Continuum spectra, Splitting and Broadening of spectral lines, polarisation

01.05 Nuclear Physics

Basic concepts including structure of atom, Mass defect and binding energy, Radioactivity, Neutrinos (Q)

COORDINATES AND TIMES

02.01 Celestial Sphere

Spherical trigonometry, Celestial coordinates and their applications, Equinox and Solstice, Circumpolar stars, Constellations and Zodiac (Note: azimuth is measured in the range 0° to 360° starting from N and increasing towards E unless stated otherwise.)

02.02 Concept of Time

Solar time, Sidereal time, Julian date, Heliocentric Julian date, Time zone, Universal Time, Local Mean Time, Different definitions of “year”, Equation of time

SOLAR SYSTEM

03.01 The Sun

Solar structure, Solar surface activities, Solar rotation, Solar radiation and Solar constant, Solar neutrinos (Q), Sun-Earth relations, Role of magnetic fields (Q), Solar wind and radiation pressure, Heliosphere (Q), Magnetosphere (Q)

03.02 The Solar System

Earth-Moon System, precession, nutation, libration, Formation and evolution of the Solar System (Q), Structure and components of the Solar System (Q), Structure and orbits of the Solar System objects, Sidereal and Synodic periods, Retrograde motion, Outer reaches of the solar system (Q)

03.03 Space Exploration

Satellite trajectories and transfers, Human exploration of the Solar System (Q), planetary missions (Q), Sling-shot effect of gravity, Space-based instruments (Q)

03.04 Phenomena

Tides, Seasons, Factors influencing climate (Q), Eclipses, Aurorae and space weather (Q), Meteor Showers

STARS

04.01 Stellar Properties

Methods of Distance determination, Radiation, Luminosity and magnitude, Color indices and temperature, Determination of radii and masses, Stellar motion, Irregular and regular stellar variabilities – broad classification & properties, Cepheids and period-luminosity relation, Physics of pulsation (Q)

04.02 Stellar Interior and Atmospheres

Stellar equilibrium, Stellar nucleosynthesis, Energy transportation (Q), Boundary conditions, Stellar atmospheres and atmospheric spectra

04.03 Stellar Evolution

Stellar formation, Hertzsprung-Russell diagram, Pre-Main Sequence, Main Sequence, Post-Main Sequence stars, supernovae, planetary nebulae, End states of stars

STELLAR SYSTEMS

05.01 Binary Star Systems

Different types of binary stars, Mass determination in binary star systems, Light and radial velocity curves of eclipsing binary systems, Doppler shifts in binary systems, interacting binaries, peculiar binary systems

05.02 Exoplanets

Techniques used to detect exoplanets, Habitable zone, Classes of exoplanets (Q), Spectral signatures of possible life (Q)

05.03 Star Clusters

Classification and Structure, Mass, age, luminosity and distance determination

05.04 Milky Way Galaxy

Structure and composition, Rotation, Satellites of Milky Way (Q)

05.05 Interstellar Medium

Gas (Q), dust (Q), HII regions, 21cm radiation, nebulae (Q), interstellar absorption, dispersion measure, Faraday rotation

05.06 Galaxies

Classifications based on structure, composition and activity, Mass, luminosity and distance determination, Rotation curves

05.07 Accretion Processes

Basic concepts (spherical and disc accretion) (Q), Eddington luminosity

COSMOLOGY

06.01 Elementary Cosmology

Expanding Universe and Hubble’s Law, Cluster of galaxies, Dark matter, Dark energy (Q), Gravitational lensing, Cosmic Microwave Background Radiation, Big Bang (Q), Alternative models of the Universe (Q), Large scale structure (Q), Distance measurement at cosmological scale, cosmological redshift

INSTRUMENTATION AND SPACE TECHNOLOGIES

07.01 Multi-wavelength Astronomy

Observations in radio, microwave, infrared, visible, ultraviolet, X-ray, and gamma-ray wavelength bands, Earth’s atmospheric effects, Artificial light and EM pollution

07.02 Instrumentation

Telescopes and detectors (e.g. charge-coupled devices, photometers, spectrographs), Magnification, Focal length, Focal ratio, resolving and light-gathering powers of telescopes, Geometric model of two element interferometer, Aperture synthesis, Adaptive optics, photometry, astrometry

MATHEMATICAL METHODS AND TOOLS

08.01 Numerical Methods

Linearisation of equations and expressions, Iterative solving of functions, estimating area under a curve by graphical method and/or by integration and/or by numerical approximations, Taylor series approximations of common functions

08.02 Basic Calculus

Derivatives of elementary functions, their sums, products, quotients, and nested functions. Integration as the inverse procedure to differentiation. Finding definite and indefinite integrals for elementary functions, and sums of functions. Geometric interpretation of derivatives and integrals. Finding constants of integration using initial conditions.

08.03 Vectors

Basic properties of vector sums, dot and cross products, Geometrical interpretation of a time derivative of a vector quantity

08.04 Geometric Instruments

Use of ruler, geometric compass, protractor, set-square

08.05 Statistics & Error Analysis

Mean, Median, Mode, percentiles, box plots, Standard deviation, basic probabilities, relative errors, error estimation using maximum error and/or standard error

PRACTICAL PART

This part consists of observations and data-analysis sections. The theoretical part of the Syllabus provides the basis for all practical problems.

The observations section focuses on contestants' experience in:

PRACTICAL PART — OBSERVATIONS

09.01 Naked-Eye Observations

Naked-Eye Observations

09.02 Sky Maps and Catalogues

Usage of sky maps and catalogues (note: any stars referred to by name rather than Bayer designation or catalogue number must be on the list of IAU-approved star names; knowledge of the whole list should not be required).

09.03 Observational Coordinates and Estimates

Application of coordinate systems in the sky, magnitude estimation, estimation of angular separation.

09.04 Astronomical Instruments

Usage of basic astronomical instruments-telescopes and various detectors for observations but enough instructions must be provided to the contestants.

Observational objects may be real sources in the sky or imitated sources in the laboratory.

Computer simulations may be used, with sufficient instructions provided to contestants.

The data-analysis section focuses on calculation and analysis of astronomical data provided in the problems.

PRACTICAL PART — DATA ANALYSIS

10.01 Error Sources

Proper identification of error sources

10.02 Graphs and Best Fit

Proper use of graph papers with different scales, e.g., polar and logarithmic papers. Finding the “Best Fit” line approximately.

10.03 Geometrical Instruments in Practice

Familiarity with the use of basic geometrical instruments (ruler, compass, protractor, set-square).

10.04 Experimental Techniques

Knowledge of the most common experimental techniques for measuring physical quantities mentioned in Part A.